Pasidu Sulochana

Mechanical Engineer

Phone: 0770075603 — Email: kpsulochana2@gmail.com — LinkedIn Profile

Professional Summary

Motivated Mechanical Engineering undergraduate with strong foundations in mechanical design, simulation, manufacturing systems, and automation. Experienced in CAD modeling, engineering analysis, PLC programming, and process improvement through academic projects and industrial training. Skilled in applying engineering principles to develop practical, efficient, and manufacturable solutions.

Experience

Assistant Engineer (Intern) Dec 2024 – May 2025
Adamjee Lukmanjee Exports Pvt Ltd

- Assisted in maintenance, troubleshooting, and performance monitoring of mechanical systems in a manufacturing environment.
- Supported process improvement activities to reduce downtime and improve operational efficiency.
- Worked with senior engineers on equipment inspection, technical documentation, and maintenance planning.

Education

BSc (Hons) in Mechanical Engineering Expected 2026
University of Moratuwa
CGPA: 3.81

GCE Advanced Level Examination – Physical Science Stream 2020
Central College Kuliyapitiya

- Results: 3A's
- Z-score: 2.3255

Technical Skills

CAD & Modeling: SolidWorks, Solid Edge, AutoCAD
Simulation Tools: ANSYS, Rocky DEM, FluidSim, Technomatix Plant Simulation
Automation & Control: PLC Programming using SIMATIC STEP 7, NI LabVIEW
Programming: C++, MATLAB, Python
Electronics/Embedded: Proteus
Languages: English, Sinhala (Native)

Projects

Development of a Domestic Food Waste Dryer with Grinding and Pelletizing Units

- Designed and modeled an integrated system combining drying, grinding, and extrusion-based pelletizing.
- Performed basic thermal analysis for moisture reduction and evaluated drying efficiency.
- Developed screw and kneader configurations for effective mixing and pellet formation.
- Considered material selection, manufacturability, and energy consumption.

Automated Wastewater Reclamation System for Cafeterias

- Designed a treatment system for greywater reuse in non-potable applications.
- Incorporated sensors and control logic for automated operation and monitoring.

Reverse Engineering Project

- Measured and analyzed existing mechanical components of a toaster to recreate accurate CAD models.
- Conducted analysis to understand design intent suggest improvements.
- Produced detailed engineering drawings for manufacturing and manufacturing methods were prposed.

Gearbox Design for Tractor PTO Shafts

- Designed gears and shifter mechanism considering torque transmission, bending stress, fatigue and design considerations.
- Performed detailed calculations for gear ratios, power requirements, and failures.
- Selected suitable materials and accounted for lubrication and wear.

Analysis of Ice Cube Manufacturing Process

- Analyzed refrigeration cycle and heat transfer processes involved in ice production.
- Identified inefficiencies and proposed improvements to enhance thermal performance.
- Evaluated system parameters such as freezing time and energy consumption.

Reverse Engineering of Coconut Deshelling Blades and Expeller Worm

- Studied geometry and wear characteristics of coconut deshelling blades.
- Modeled components in CAD and analyzed their functional performance.
- Suggested design improvements for durability and efficiency and manufactured the blades as per requirements of the production.

Design of a Pineapple Slicing Machine for Production Line

- Designed a mechanical slicing machine to improve uniformity and throughput.
- Developed concepts considering ergonomics, safety, and continuous operation.
- Created CAD models and simulations for motion analysis and stress analysis for mechanism validation.

Leadership & Engagement

- Member of ASHRAE Student Branch - University of Moratuwa
- Member of IMechE Student Branch - University of Moratuwa

References

Dr. Subasinghe LU
Senior Lecturer
Department of Mechanical Engineering
University of Moratuwa
Sri Lanka
Email: lihils@uom.lk